Mental Health and Crime Rate

Two Independent of the Urban Landscape or Potentially Something Deeper?

DSCI 510

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Introduction

WHAT

This project will explore the relationship between **mental health** status and **crime rates** in the United States with socioeconomic confounders at the **state-level**.

WHY

- useful to many policymakers and think tanks
- explores whether investing funds and resources into mental health programs, shelters, and aid would improve the quality of life for residents through the reduced crime

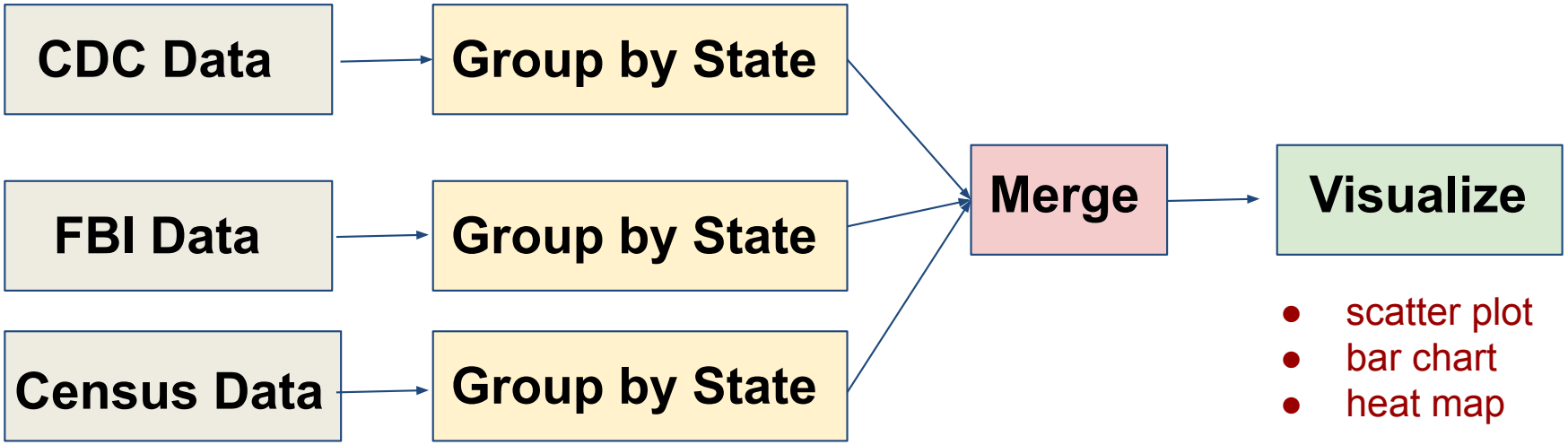
HOW

- use publicly available data from the CDC, FBI, and the US Census Bureau
- explore the relationship between mental health responses and crime rates, analyzing correlation coefficients and visualization patterns
- help us better understand, alongside some confounding variables, whether mental health and crime are related in *any* way

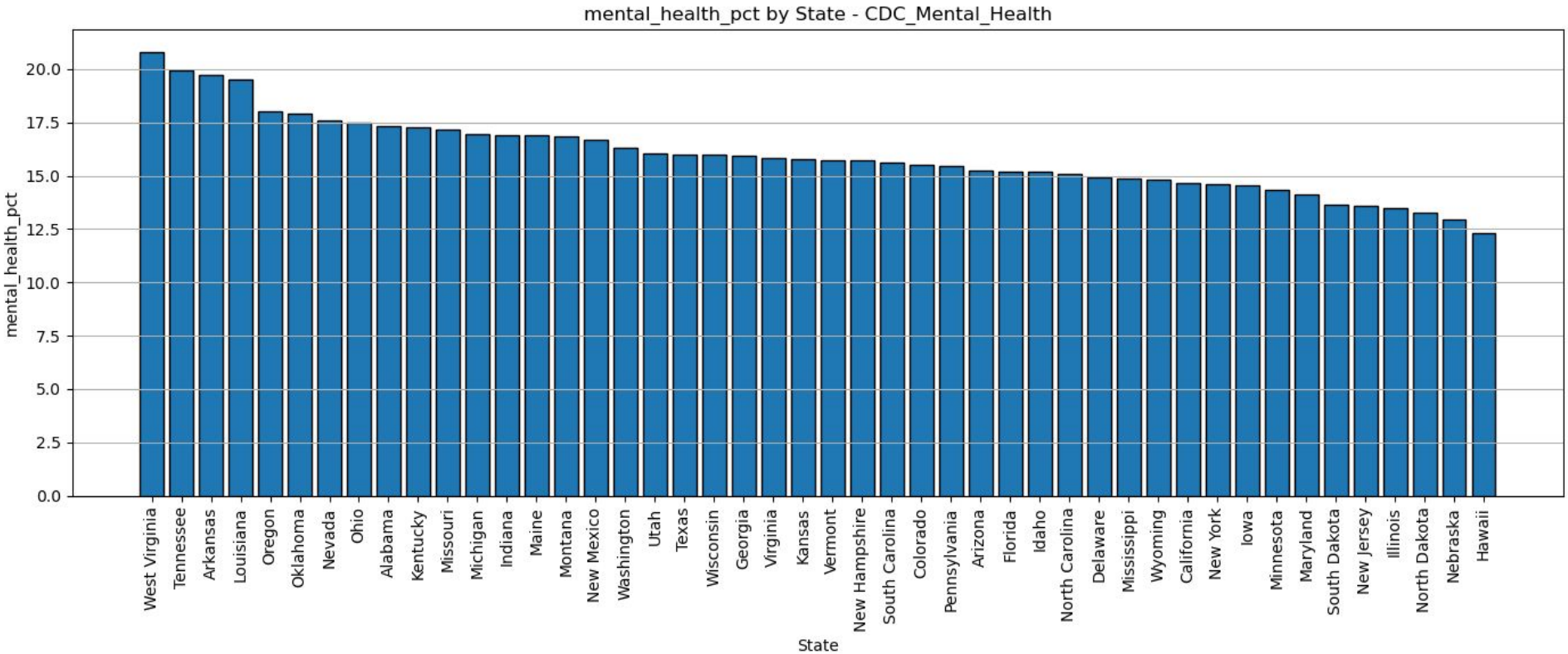
Data Sources

Source	Name/Description	Format	Important Features
CDC (Center for Disease Control and Prevention)	~ Behavioral Risk Factor Surveillance System (BRFSS) - Mental Health Indicators ~ Mental health survey data in the United States	API JSON	Now thinking about your mental health, which includes stress, depression, and problems with emotions, for how many days during the past 30 days was your mental health not good?
FBI's Crime Data Explorer	~Crime in the United States Annual Reports - 2024, Offenses Known to Law Enforcement ~ Crime data categorized by offense and state	CSV	Violent Crime
United States Census Bureau	~ ACS 5-year Estimates (2020-2025), Selected Economic Characteristics ~ Economic demographic data for counties in the US	CSV	'Geographic Area Name', 'Percent!!PERCENTAGE OF FAMILIES AND PEOPLE WHOSE INCOME IN THE PAST 12 MONTHS IS BELOW THE POVERTY LEVEL!!All people', 'Percent!!HEALTH INSURANCE COVERAGE!!Civilian noninstitutionalized population!!No health insurance coverage', 'Percent!!EMPLOYMENT STATUS!!Civilian labor force!!Unemployment Rate'

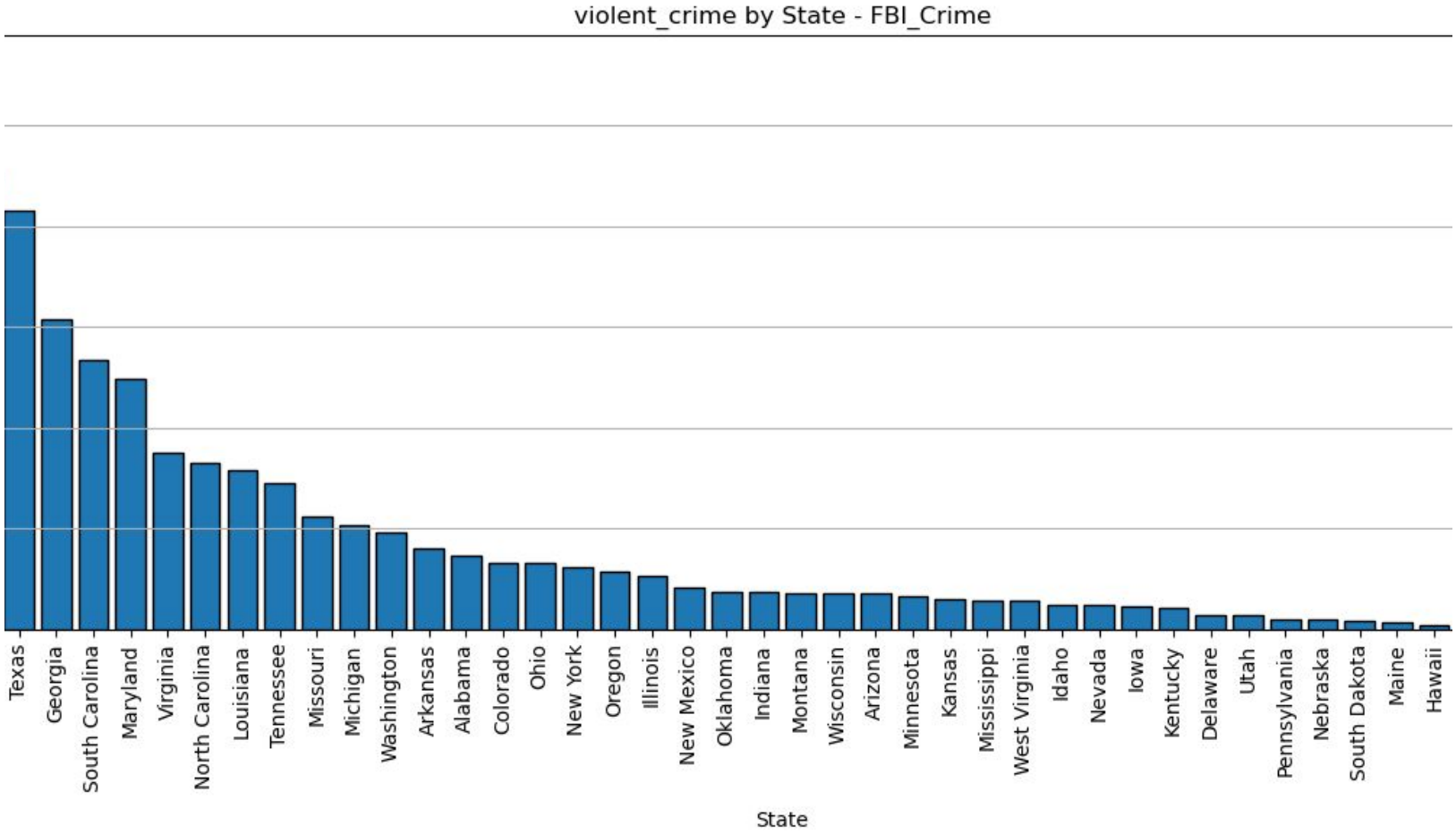
Methods


CDC Data
FBI Data
Census Data
Group by State
Group by State
Group by State
Merge
Visualize
scatter plot
bar chart
heat map

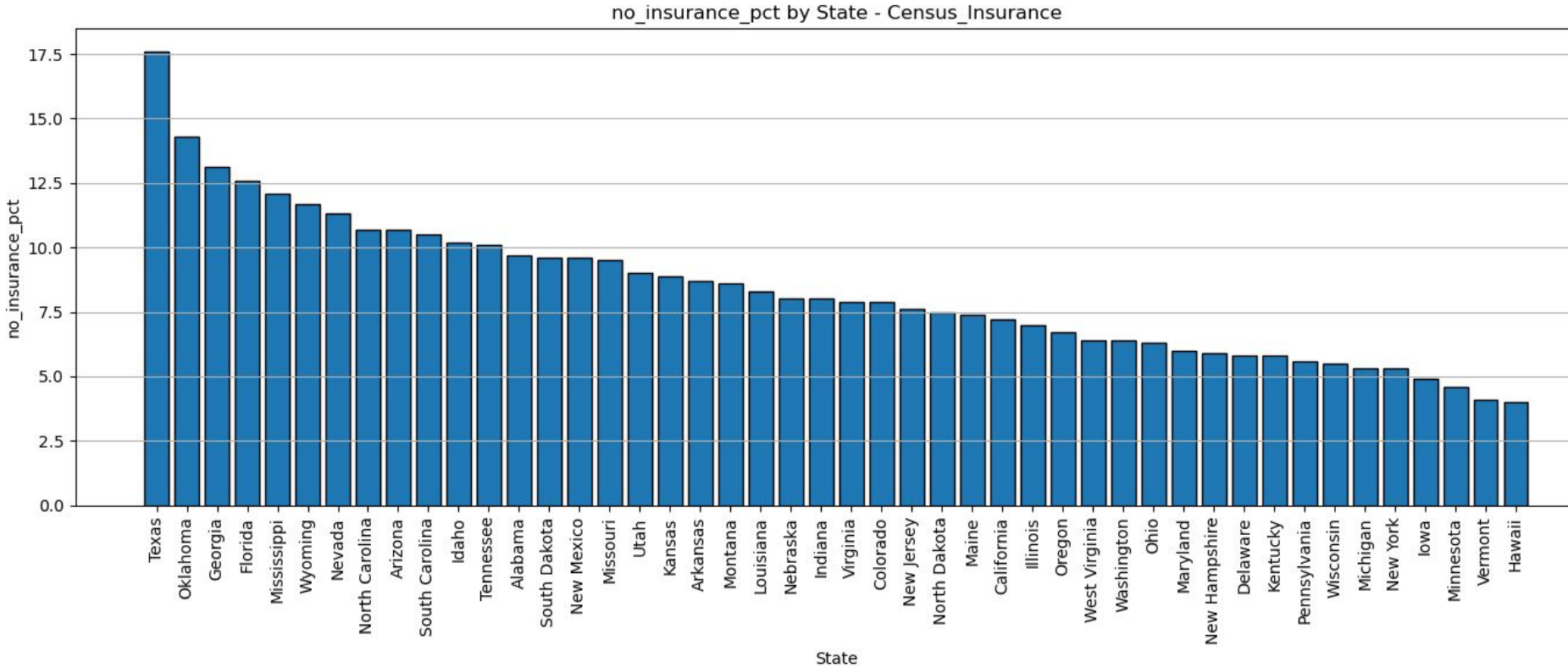
Results


mental_health_pct by State - CDC_Mental_Health
mental_health_pct
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
West Virginia
Tennessee
Arkansas
Louisiana
Oregon
Oklahoma
Nevada
Ohio
Alabama
Kentucky
Missouri
Michigan
Indiana
Maine
Montana
New Mexico
Washington
Utah
Texas
Wisconsin
Georgia
Virginia
Kansas
Vermont
New Hampshire
South Carolina
Colorado
Pennsylvania
Arizona
Florida
Idaho
North Carolina
Delaware
Mississippi
Wyoming
California
New York
Iowa
Minnesota
Maryland
South Dakota
New Jersey
Illinois
North Dakota
Nebraska
Hawaii
State

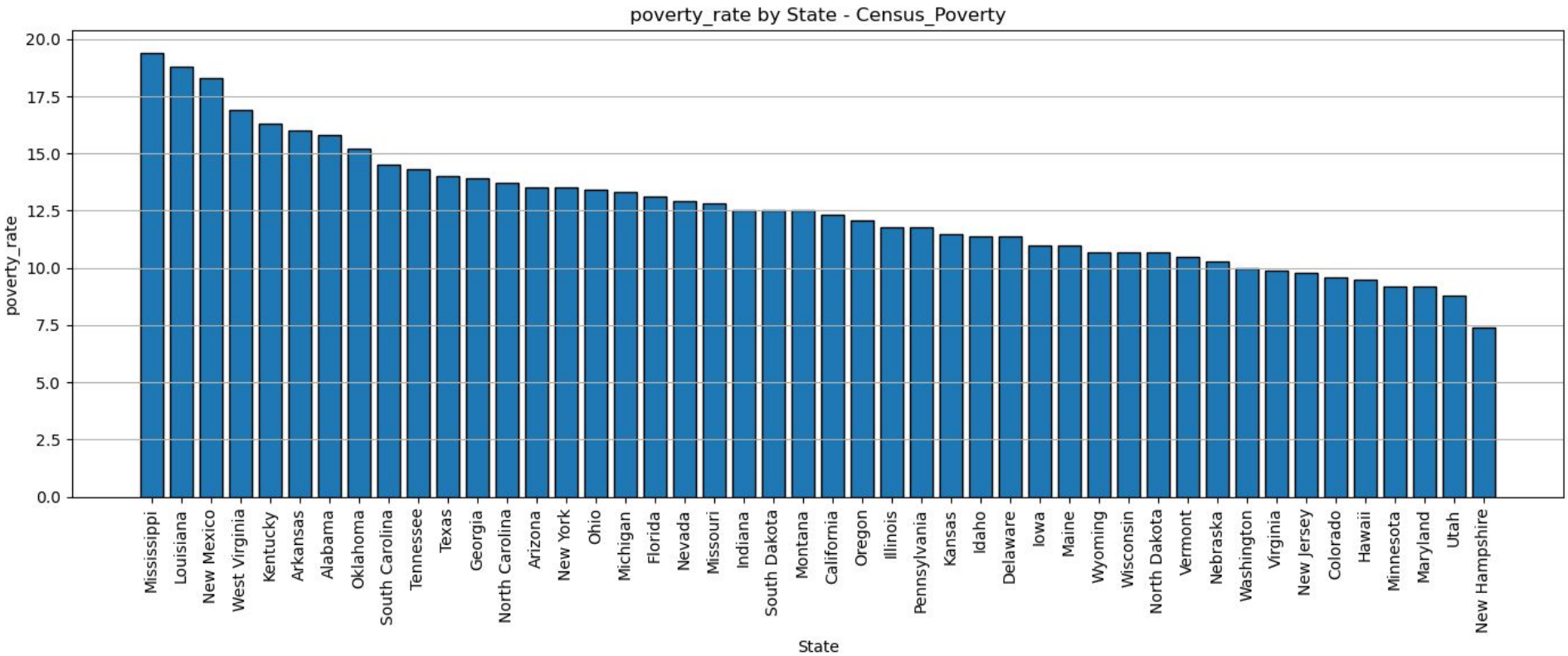
Results


violent_crime by State - FBI_Crime
Texas
Georgia
South Carolina
Maryland
Virginia
North Carolina
Louisiana
Tennessee
Missouri
Michigan
Washington
Arkansas
Alabama
Colorado
Ohio
New York
Oregon
Illinois
New Mexico
Oklahoma
Indiana
Montana
Wisconsin
Arizona
Minnesota
Kansas
Mississippi
West Virginia
Idaho
Nevada
Iowa
Kentucky
Delaware
Utah
Pennsylvania
Nebraska
South Dakota
Maine
Hawaii
State

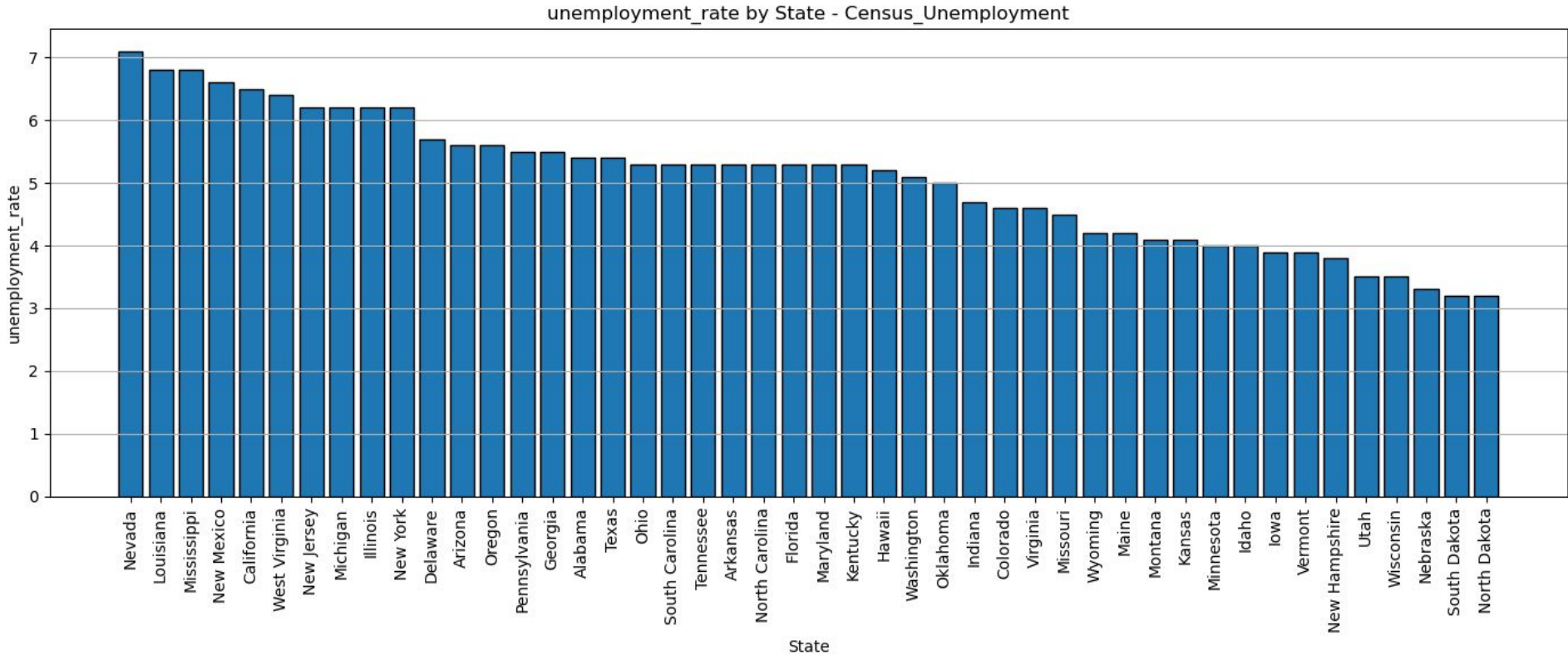
Results


no_insurance_pct by State - Census_Insurance
no_insurance_pct
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
Texas
Oklahoma
Georgia
Florida
Mississippi
Wyoming
Nevada
North Carolina
Arizona
South Carolina
Idaho
Tennessee
Alabama
South Dakota
New Mexico
Missouri
Utah
Kansas
Arkansas
Montana
Louisiana
Nebraska
Indiana
Virginia
Colorado
New Jersey
North Dakota
Maine
California
Illinois
Oregon
West Virginia
Washington
Ohio
Maryland
New Hampshire
Delaware
Kentucky
Pennsylvania
Wisconsin
Michigan
New York
Iowa
Minnesota
Vermont
Hawaii
State

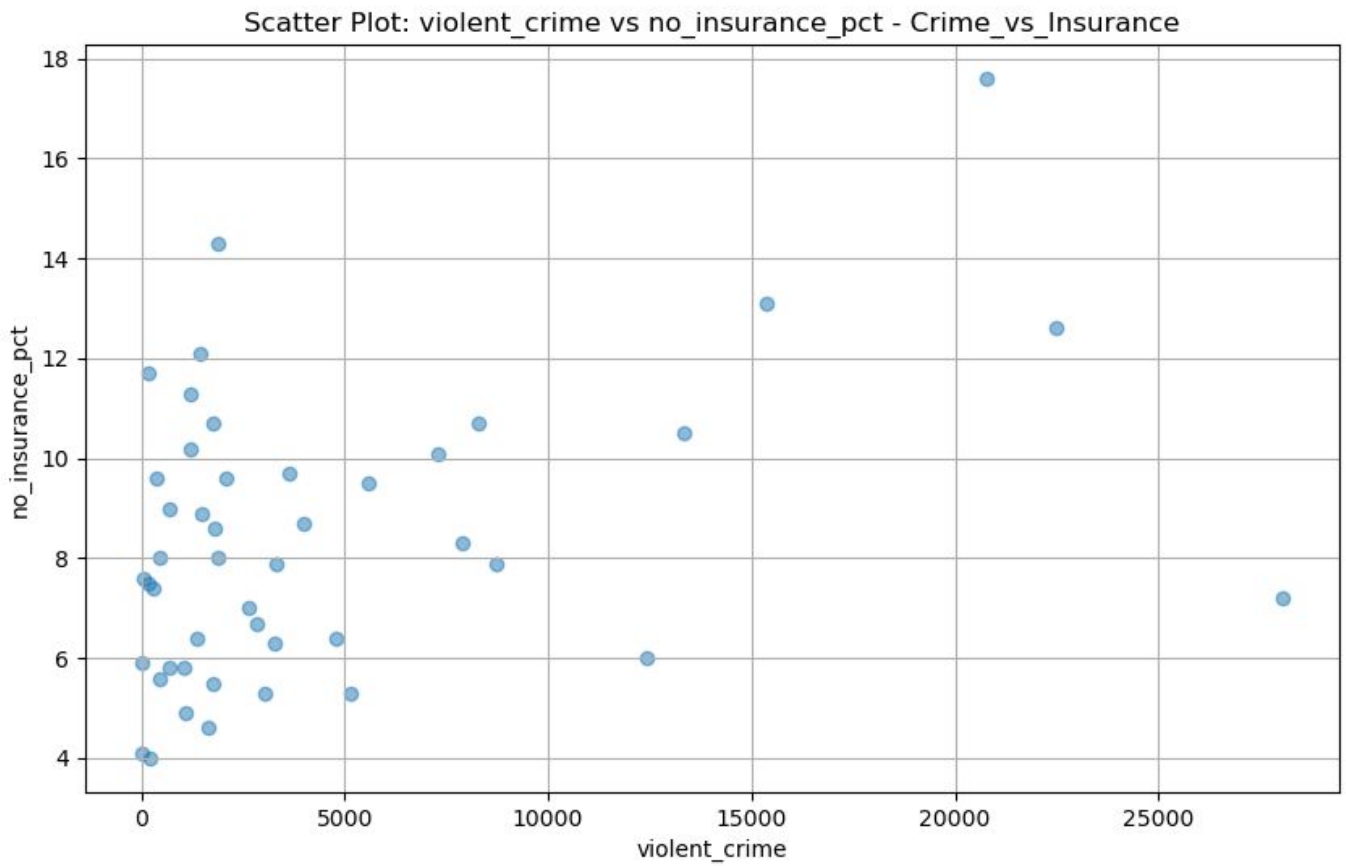
Results


poverty_rate by State - Census_Poverty
poverty_rate
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
Mississippi
Louisiana
New Mexico
West Virginia
Kentucky
Arkansas
Alabama
Oklahoma
South Carolina
Tennessee
Texas
Georgia
North Carolina
Arizona
New York
Ohio
Michigan
Florida
Nevada
Missouri
Indiana
South Dakota
Montana
California
Oregon
Illinois
Pennsylvania
Kansas
Idaho
Delaware
Iowa
Maine
Wyoming
Wisconsin
North Dakota
Vermont
Nebraska
Washington
Virginia
New Jersey
Colorado
Hawaii
Minnesota
Maryland
Utah
New Hampshire
State

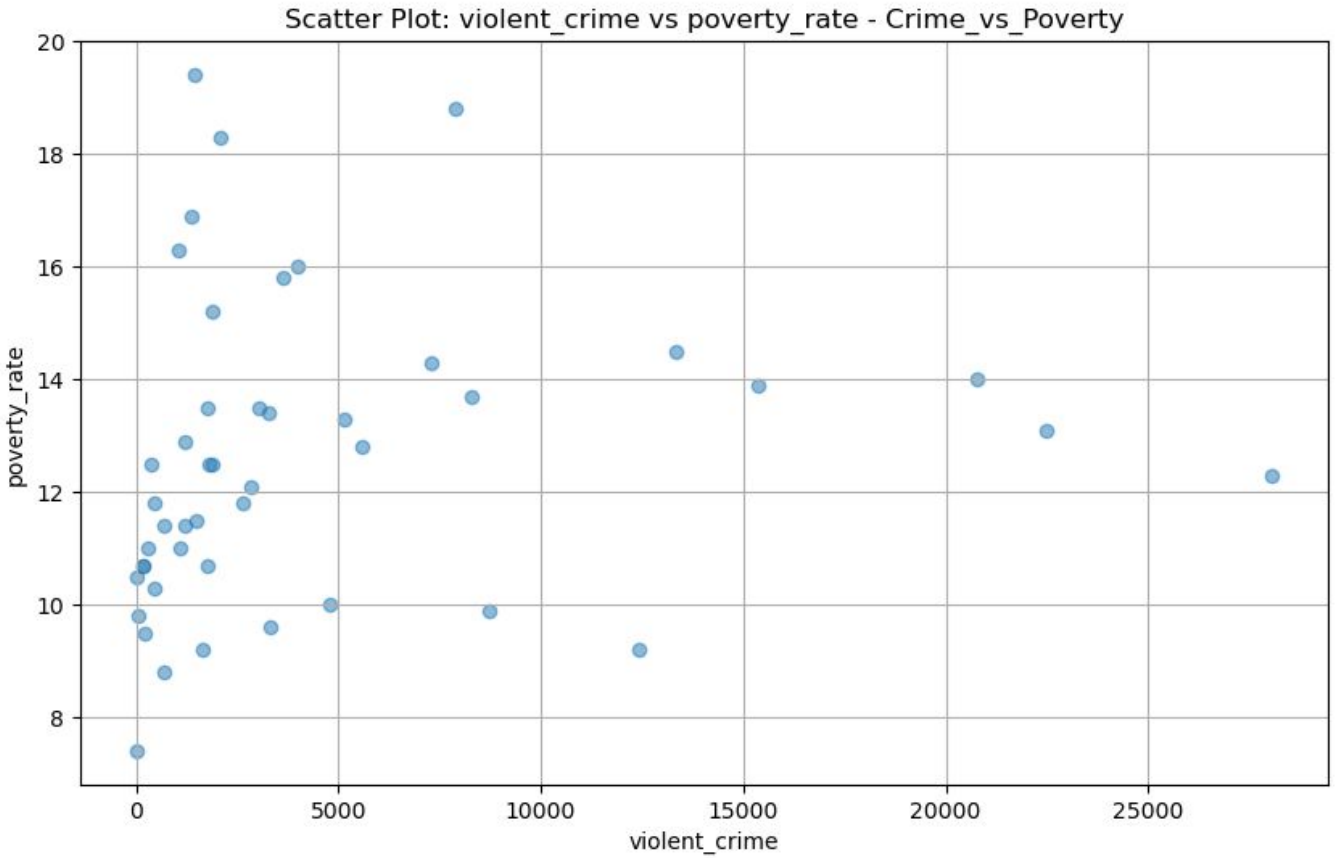
Results


unemployment_rate by State - Census_Unemployment
unemployment_rate
0
1
2
3
4
5
6
7
Nevada
Louisiana
Mississippi
New Mexico
California
West Virginia
New Jersey
Michigan
Illinois
New York
Delaware
Arizona
Oregon
Pennsylvania
Georgia
Alabama
Texas
Ohio
South Carolina
Tennessee
Arkansas
North Carolina
Florida
Maryland
Kentucky
Hawaii
Washington
Oklahoma
Indiana
Colorado
Virginia
Missouri
Wyoming
Maine
Montana
Kansas
Minnesota
Idaho
Iowa
Vermont
New Hampshire
Utah
Wisconsin
Nebraska
South Dakota
North Dakota
State

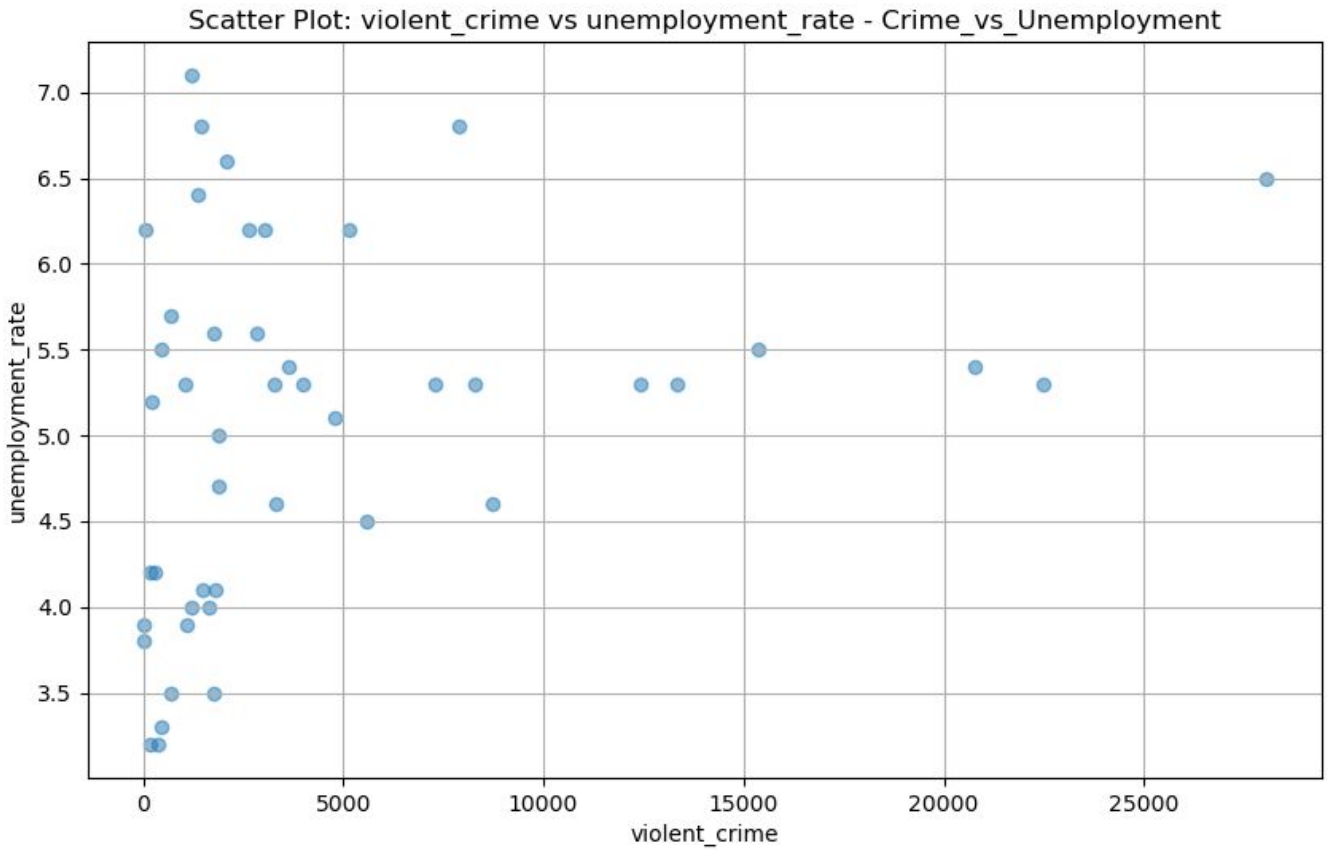
Results


Scatter Plot: violent_crime vs no_insurance_pct - Crime_vs_Insurance
no_insurance_pct
violent_crime
18
16
14
12
10
8
6
4
0
5000
10000
15000
20000
25000

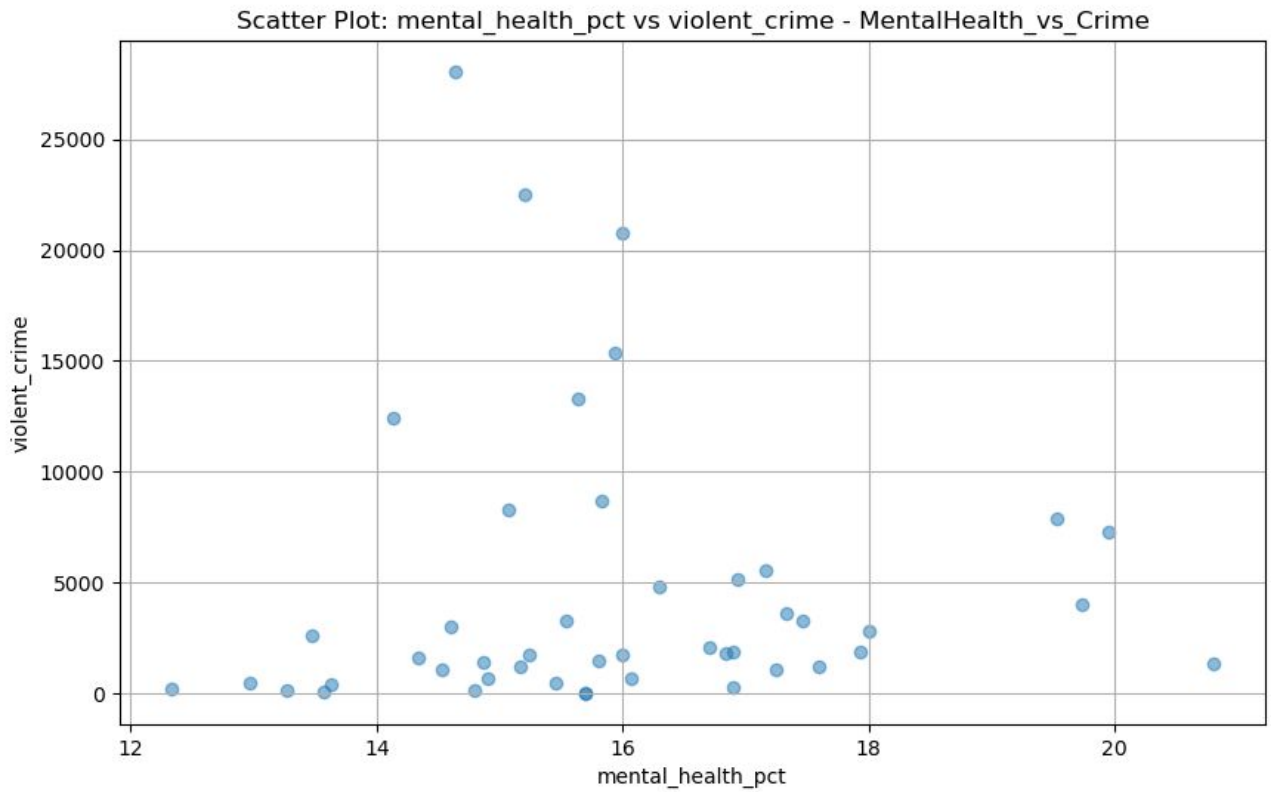
Results


Scatter Plot: violent_crime vs poverty_rate - Crime_vs_Poverty
poverty_rate
violent_crime
20
18
16
14
12
10
8
0
5000
10000
15000
20000
25000

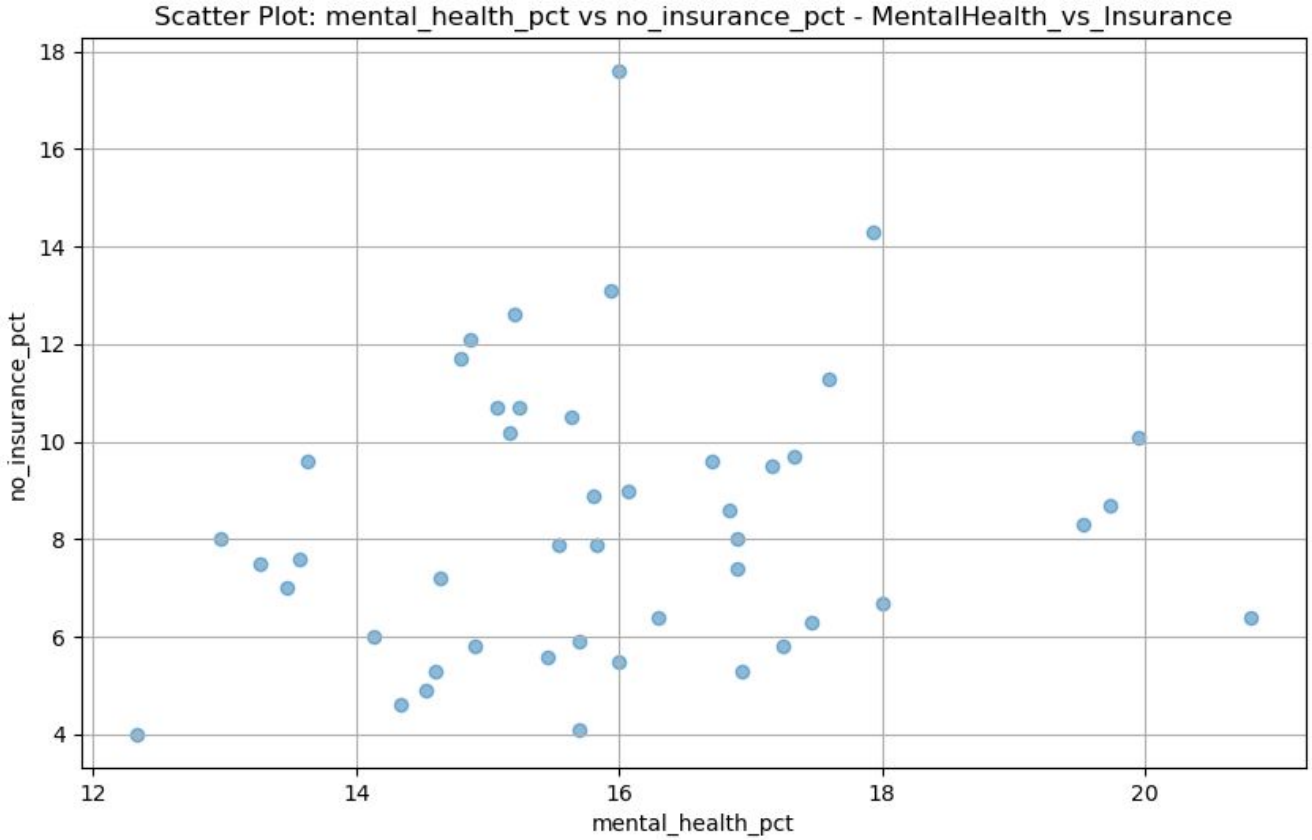
Results


Scatter Plot: violent_crime vs unemployment_rate - Crime_vs_Unemployment
unemployment_rate
violent_crime
7.0
6.5
6.0
5.5
5.0
4.5
4.0
3.5
0
5000
10000
15000
20000
25000

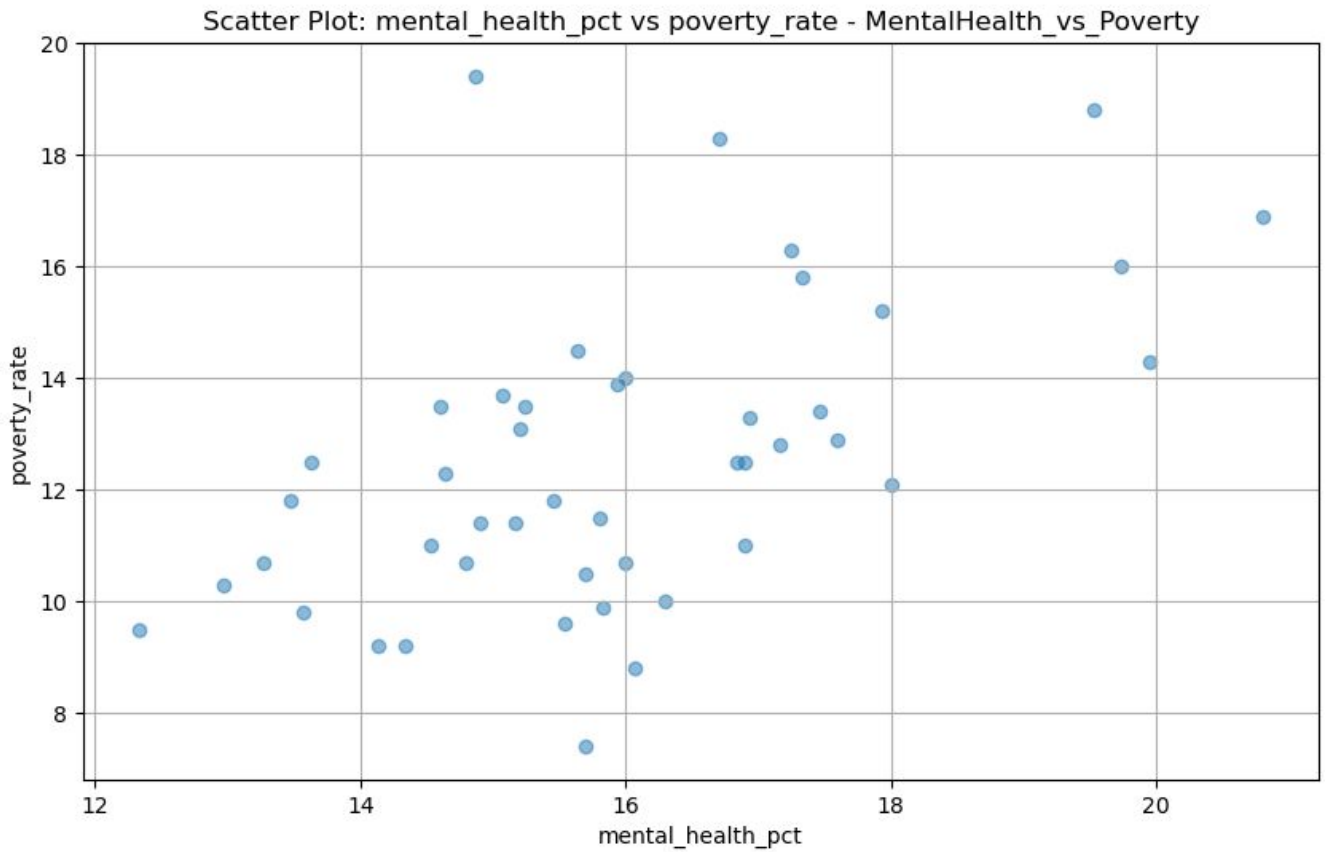
Results


Scatter Plot: mental_health_pct vs violent_crime - MentalHealth_vs_Crime
violent_crime
25000
20000
15000
10000
5000
0
12
14
16
18
20
mental_health_pct

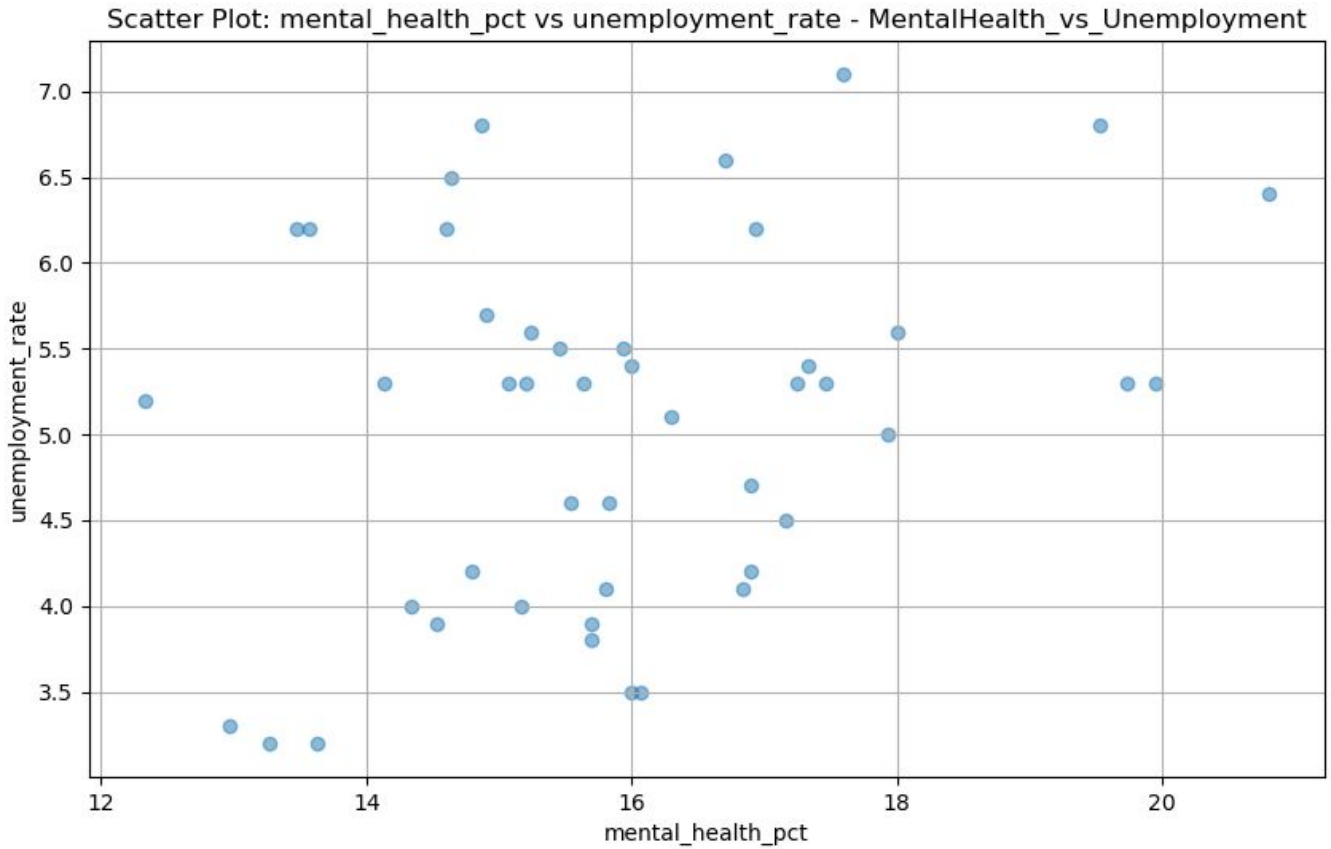
Results


Scatter Plot: mental_health_pct vs no_insurance_pct - MentalHealth_vs_Insurance
no_insurance_pct
mental_health_pct
18
16
14
12
10
8
6
4
12
14
16
18
20

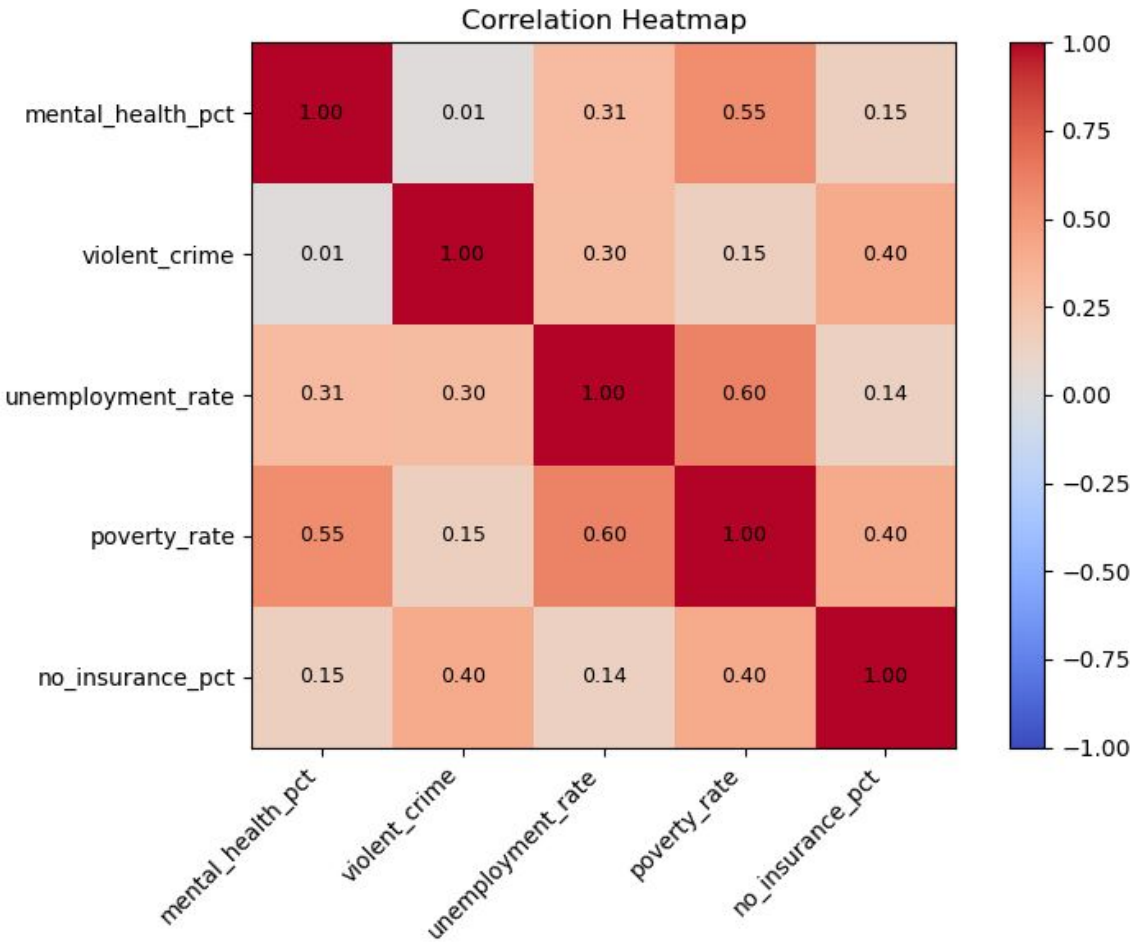
Results


Scatter Plot: mental_health_pct vs poverty_rate - MentalHealth_vs_Poverty
poverty_rate
mental_health_pct
20
18
16
14
12
10
8
12
14
16
18
20

Results


Scatter Plot: mental_health_pct vs unemployment_rate - MentalHealth_vs_Unemployment
unemployment_rate
7.0
6.5
6.0
5.5
5.0
4.5
4.0
3.5
12
14
16
18
20
mental_health_pct

Results


Correlation Heatmap
mental_health_pct
violent_crime
unemployment_rate
poverty_rate
no_insurance_pct
1.00 0.01 0.31 0.55 0.15
0.01 1.00 0.30 0.15 0.40
0.31 0.30 1.00 0.60 0.14
0.55 0.15 0.60 1.00 0.40
0.15 0.40 0.14 0.40 1.00
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

Challenges

- Cleaning data to prepare for merge
- Keeping track of all 6 factors
- Working with pandas
- Loading DFs

- missing states
- confounding variables
- may need to run a full regression

Thank You!